ISE 5405: Optimization I

Lecture 6: Simplex I
From a Basis to a Better Corner

BT Sections 3.1–3.2

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What You Should Be Able to Do

1. Explain why simplex moves between basic feasible solutions.
2. Construct a basic direction by solving one basis system.
3. Interpret a reduced cost as an objective slope.
4. Apply the ratio test and identify an entering and leaving variable.
5. Recognize optimality, unboundedness, and a degenerate pivot.

Today's route

See the path, derive the direction, price the direction, then take one pivot.

The Standard-Form Setting

We study

$$\min\{c^\top x : Ax = b,\ x \geq 0\}, \qquad A \in \mathbb{R}^{m \times n}, \quad \operatorname{rank}(A) = m.$$

A feasible basis chooses m independent columns $B = A_{\mathcal{B}}$ and sets

$$x_{\mathcal{B}} = B^{-1} b \geq 0, \qquad x_{\mathcal{N}} = 0.$$

Starting point

Primal simplex starts from a feasible basis and preserves feasibility while it searches for a lower objective value.

The Simplex Method: The Big Picture

1. Start at a basic feasible solution, geometrically a corner.
2. Look along incident basic directions for a cost-reducing direction.
3. Move as far as feasibility permits to a neighboring corner.
4. Stop when no nonbasic direction has a negative reduced cost.

Why corners are enough

If a linear program has an optimal solution, then it has an optimal basic feasible solution.

How to Read the GILP Sequence

The GILP reference sequence uses

$$\max\ z = 5x_1 + 3x_2,$$

whereas our algebra uses minimization. Multiplying the objective by -1 connects the conventions:

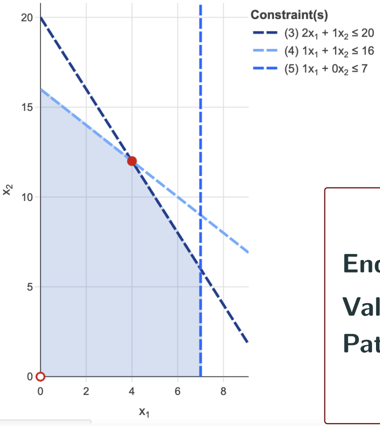
$$\text{larger } z \text{ in the picture}$$
$$\Updownarrow$$
$$\text{negative reduced cost for minimizing } -z$$

Watch three things

The red path accumulates, the current basis dictionary changes, and every move follows an edge of the blue feasible set.

All Integer 2D LP

A 2D LP where all basic feasible solutions are integral and have integral tableaus.



(0)

$$\begin{array}{lll} \text{max} & z & = 0 + 5x_1 + 3x_2 \\ \text{s.t.} & x_3 & = 20 - 2x_1 - 1x_2 \\ & x_4 & = 16 - 1x_1 - 1x_2 \\ & x_5 & = 7 - 1x_1 + 0x_2 \end{array}$$

Endpoint: $(0,0)$

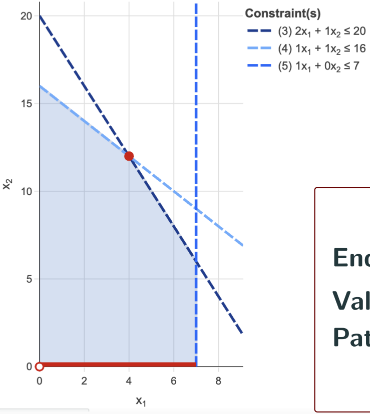
Value: $z = 0$

Path: $(0,0)$

GILP Path in Two Dimensions: Iteration 1 at $(7,0)$

All Integer 2D LP

A 2D LP where all basic feasible solutions are integral and have integral tableaus.



(1)

$$\begin{array}{lll} \text{max} & z & = 35 + 3x_2 - 5x_5 \\ \text{s.t.} & x_1 & = 7 + 0x_2 - 1x_5 \\ & x_3 & = 6 - 1x_2 + 2x_5 \\ & x_4 & = 9 - 1x_2 + 1x_5 \end{array}$$

Endpoint: $(7,0)$

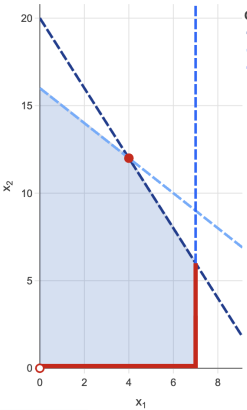
Value: $z = 35$

Path: $(0,0) \rightarrow (7,0)$

GILP Path in Two Dimensions: Iteration 2 at $(7, 6)$

All Integer 2D LP

A 2D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

- (3) $2x_1 + 1x_2 \leq 20$
- (4) $1x_1 + 1x_2 \leq 16$
- (5) $1x_1 + 0x_2 \leq 7$

(2)

$$
\begin{array}{lll}
\max & z & = 53 - 3x_3 + 1x_5 \\
\text{s.t.} & x_1 & = 7 + 0x_3 - 1x_5 \\
 & x_2 & = 6 - 1x_3 + 2x_5 \\
 & x_4 & = 3 + 1x_3 - 1x_5
\end{array}
$$

Endpoint: $(7, 6)$

Value: $z = 53$

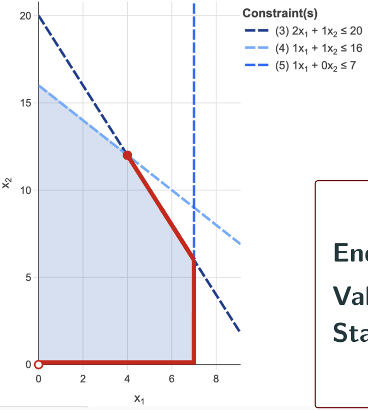
Path:

$(0,0) \to (7,0) \to (7,6)$

GILP Path in Two Dimensions: Iteration 3 at $(4, 12)$

All Integer 2D LP

A 2D LP where all basic feasible solutions are integral and have integral tableaus.



(3)

$$
\begin{array}{lll}
\max & z & = 56 - 2x_3 - 1x_4 \\
\text{s.t.} & x_1 & = 4 - 1x_3 + 1x_4 \\
 & x_2 & = 12 + 1x_3 - 2x_4 \\
 & x_5 & = 3 + 1x_3 - 1x_4
\end{array}
$$

Endpoint: $(4, 12)$

Value: $z = 56$

Status: optimal

What the 2D Path Shows

$$(0, 0) \longrightarrow (7, 0) \longrightarrow (7, 6) \longrightarrow (4, 12)$$

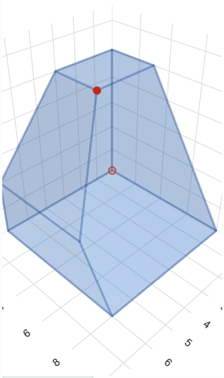
$$z : 0 \longrightarrow 35 \longrightarrow 53 \longrightarrow 56.$$

- Each red segment stays feasible and connects neighboring corners.
- Each dictionary expresses the objective and basic variables in terms of the current nonbasic variables.
- At the last corner, no available edge can increase z.

GILP Path in Three Dimensions: Iteration 0

All Integer 3D LP

A 3D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

(4) $1x_1 + 0x_2 + 0x_3 \leq 6$

(5) $1x_1 + 0x_2 + 1x_3 \leq 8$

(6) $0x_1 + 0x_2 + 1x_3 \leq 5$

(7) $0x_1 + 1x_2 + 1x_3 \leq 8$

(0)

$$
\begin{array}{lll}
\text{max} & z & = 0 + 1x_1 + 2x_2 + 4x_3 \\
\text{s.t.} & x_4 & = 6 - 1x_1 + 0x_2 + 0x_3 \\
 & x_5 & = 8 - 1x_1 + 0x_2 - 1x_3 \\
 & x_6 & = 5 + 0x_1 + 0x_2 - 1x_3 \\
 & x_7 & = 8 + 0x_1 - 1x_2 - 1x_3
\end{array}
$$

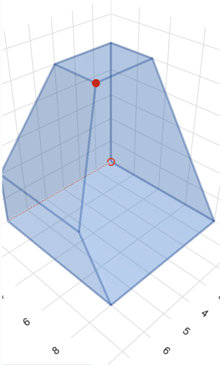
Endpoint: $(0, 0, 0)$

Value: $z = 0$

Path: origin

All Integer 3D LP

A 3D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

(4) $1x_1 + 0x_2 + 0x_3 \leq 6$

(5) $1x_1 + 0x_2 + 1x_3 \leq 8$

(6) $0x_1 + 0x_2 + 1x_3 \leq 5$

(7) $0x_1 + 1x_2 + 1x_3 \leq 8$

(1)

$$
\begin{array}{lll}
\text{max} & z & = 6 + 2x_2 + 4x_3 - 1x_4 \\
\text{s.t.} & x_1 & = 6 + 0x_2 + 0x_3 - 1x_4 \\
 & x_5 & = 2 + 0x_2 - 1x_3 + 1x_4 \\
 & x_6 & = 5 + 0x_2 - 1x_3 + 0x_4 \\
 & x_7 & = 8 - 1x_2 - 1x_3 + 0x_4
\end{array}
$$

Endpoint: $(6, 0, 0)$

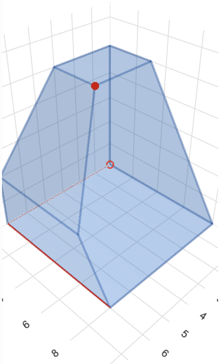
Value: $z = 6$

Step: first edge

GILP Path in Three Dimensions: Iteration 2

All Integer 3D LP

A 3D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

(4) $1x_1 + 0x_2 + 0x_3 \leq 6$
(5) $1x_1 + 0x_2 + 1x_3 \leq 8$
(6) $0x_1 + 0x_2 + 1x_3 \leq 5$
(7) $0x_1 + 1x_2 + 1x_3 \leq 8$

(2)

$$
\begin{array}{lll}
\max & z & = 22 + 2x_3 - 1x_4 - 2x_7 \\
\text{s.t.} & x_1 & = 6 + 0x_3 - 1x_4 + 0x_7 \\
 & x_2 & = 8 - 1x_3 + 0x_4 - 1x_7 \\
 & x_5 & = 2 - 1x_3 + 1x_4 + 0x_7 \\
 & x_6 & = 5 - 1x_3 + 0x_4 + 0x_7
\end{array}
$$

Endpoint: $(6, 8, 0)$

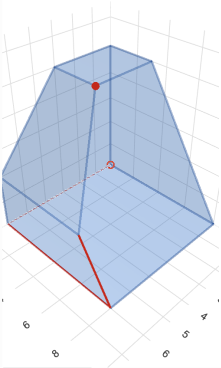
Value: $z = 22$

Step: second edge

GILP Path in Three Dimensions: Iteration 3

All Integer 3D LP

A 3D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

(4) $1x_1 + 0x_2 + 0x_3 \leq 6$
(5) $1x_1 + 0x_2 + 1x_3 \leq 8$
(6) $0x_1 + 0x_2 + 1x_3 \leq 5$
(7) $0x_1 + 1x_2 + 1x_3 \leq 8$

(3)

$$
\begin{array}{lll}
\max & z & = 26 + 1x_4 - 2x_5 - 2x_7 \\
\text{s.t.} & x_1 & = 6 - 1x_4 + 0x_5 + 0x_7 \\
 & x_2 & = 6 - 1x_4 + 1x_5 - 1x_7 \\
 & x_3 & = 2 + 1x_4 - 1x_5 + 0x_7 \\
 & x_6 & = 3 - 1x_4 + 1x_5 + 0x_7
\end{array}
$$

Endpoint: $(6, 6, 2)$

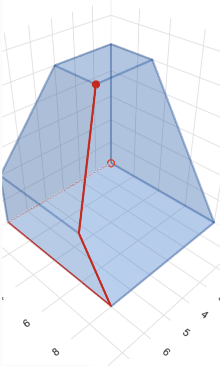
Value: $z = 26$

Step: third edge

GILP Path in Three Dimensions: Iteration 4

All Integer 3D LP

A 3D LP where all basic feasible solutions are integral and have integral tableaus.



Constraint(s)

(4) $1x_1 + 0x_2 + 0x_3 \le 6$
(5) $1x_1 + 0x_2 + 1x_3 \le 8$
(6) $0x_1 + 0x_2 + 1x_3 \le 5$
(7) $0x_1 + 1x_2 + 1x_3 \le 8$

(4)

$$
\begin{array}{lll}
\text{max} & z & = 29 - 1x_5 - 1x_6 - 2x_7 \\
\text{s.t.} & x_1 & = 3 - 1x_5 + 1x_6 + 0x_7 \\
 & x_2 & = 3 + 0x_5 + 1x_6 - 1x_7 \\
 & x_3 & = 5 + 0x_5 - 1x_6 + 0x_7 \\
 & x_4 & = 3 + 1x_5 - 1x_6 + 0x_7
\end{array}
$$

Endpoint: $(3, 3, 5)$

Value: $z = 29$

Status: final state

The Same Mechanism in 3D

- The blue polyhedron replaces the 2D polygon.
- The cumulative red path still follows one edge at a time.
- A pivot changes the basis and dictionary, not the feasible set.
- The algebra does not depend on our ability to draw the geometry.

The algorithmic question

At the current basis, which nonbasic variable gives an improving direction, and how far can we move along it?

Feasible Directions

Definition
For $x \in P$, a vector $d \in \mathbb{R}^n$ is a **feasible direction** at x if some $\epsilon > 0$ satisfies

$$x + \theta d \in P \qquad \text{for every } 0 \leq \theta \leq \epsilon.$$

For $P = \{x : Ax = b,\ x \geq 0\}$, two checks are required:

$$Ad = 0 \qquad \text{and} \qquad x + \theta d \geq 0 \text{ for a small positive step.}$$

A Basis Partitions the Variables

For a feasible basis $B = A_{\mathcal{B}}$, write

$$A = [\, B \; A_{\mathcal{N}} \,], \qquad x = \begin{bmatrix} x_{\mathcal{B}} \\ x_{\mathcal{N}} \end{bmatrix}, \qquad x_{\mathcal{B}} = B^{-1}b, \quad x_{\mathcal{N}} = 0.$$

To leave this corner, choose one nonbasic variable x_j to increase while the other nonbasic variables remain zero.

One column, one candidate edge

Each nonbasic column A_j defines one basic direction relative to the current basis.

Construct a Basic Direction: State 1 of 3

Choose a nonbasic index $j \in \mathcal{N}$. Prescribe

$$d_j = 1, \qquad d_k = 0 \quad (k \in \mathcal{N} \setminus \{j\}).$$

The unknown components are the basic components $d_{\mathcal{B}}$.

Interpretation

x_j increases at unit rate; the basic variables must compensate so that the equality constraints remain satisfied.

Construct a Basic Direction: State 2 of 3

Keep the prescribed nonbasic components visible:

$$d_j = 1, \qquad d_k = 0 \quad (k \in \mathcal{N} \setminus \{j\}).$$

Equality preservation requires

$$0 = Ad = Bd_{\mathcal{B}} + A_j.$$

Thus the basic components solve the basis system

$$Bd_{\mathcal{B}} = -A_j.$$

The complete cumulative derivation is

$$d_j = 1, \qquad d_k = 0 \ (k \in \mathcal{N} \setminus \{j\}),$$

$$0 = Ad = Bd_{\mathcal{B}} + A_j, \qquad \boxed{d_{\mathcal{B}} = -B^{-1}A_j}.$$

Compute by solving

Solve $Bd_{\mathcal{B}} = -A_j$. The inverse notation describes the answer; it is not an instruction to form B^{-1}.

Equality Preservation Is Not Enough

Nondegenerate basic feasible solution

If $x_{\mathcal{B}} > 0$, every basic direction permits a sufficiently small positive step: $x_{\mathcal{B}} + \theta d_{\mathcal{B}} \geq 0$.

Degenerate basic feasible solution

If some $x_{B(i)} = 0$ and $d_{B(i)} < 0$, then every positive step makes that component negative. The direction is blocked at $\theta = 0$.

Degeneracy separates an algebraically valid direction from a direction that permits a positive geometric move.

Objective Slope Along a Basic Direction

Along $x(\theta) = x + \theta d$,

$$c^\top x(\theta) = c^\top x + \theta c^\top d.$$

For the j-th basic direction,

$$\begin{aligned} c^\top d &= c_j + c_{\mathcal{B}}^\top d_{\mathcal{B}} \\ &= c_j - c_{\mathcal{B}}^\top B^{-1} A_j. \end{aligned}$$

This scalar is the rate of objective change per unit increase in x_j.

Reduced Cost

Definition

The reduced cost of variable x_j relative to basis B is

$$\boxed{\bar{c}_j = c_j - c_{\mathcal{B}}^{\top} B^{-1} A_j}.$$

For a minimization problem:

- $\bar{c}_j < 0$: the j-th basic direction has an improving slope;
- $\bar{c}_j = 0$: the objective is locally flat in that direction;
- $\bar{c}_j > 0$: that direction initially raises the objective.

Compute Reduced Costs with a Transpose Solve

Rather than form B^{-1}, solve

$$B^\top y = c_{\mathcal{B}}.$$

Then

$$y^\top = c_{\mathcal{B}}^\top B^{-1}, \qquad \boxed{\bar{c}_j = c_j - y^\top A_j}.$$

Reusable basis computation

The same factorization of B supports basic values, pivot columns, and the transpose solve used for pricing.

Every Basic Variable Has Zero Reduced Cost

For a basic index $B(i)$,

$$B^{-1}A_{B(i)} = e_i.$$

Therefore

$$\begin{aligned} \bar{c}_{B(i)} &= c_{B(i)} - c_{\mathcal{B}}^{\top} B^{-1} A_{B(i)} \\ &= c_{B(i)} - c_{\mathcal{B}}^{\top} e_i \\ &= 0. \end{aligned}$$

Only nonbasic reduced costs need to be examined.

Optimality Conditions

Theorem

Let x be a basic feasible solution associated with basis B.

1. *If every reduced cost is nonnegative, then x is optimal.*
2. *If x is optimal and nondegenerate, then every reduced cost is nonnegative.*

Degenerate caveat

An optimal degenerate point can have a basis with a negative reduced cost whose direction is blocked at step zero.

Why Nonnegative Reduced Costs Prove Optimality

For any feasible x,

$$x_{\mathcal{B}} = B^{-1}b - B^{-1}A_{\mathcal{N}}x_{\mathcal{N}}.$$

Substitute into the objective:

$$\begin{aligned} c^\top x &= c_{\mathcal{B}}^\top B^{-1}b + \left(c_{\mathcal{N}}^\top - c_{\mathcal{B}}^\top B^{-1}A_{\mathcal{N}}\right)x_{\mathcal{N}} \\ &= c_{\mathcal{B}}^\top B^{-1}b + \bar{c}_{\mathcal{N}}^\top x_{\mathcal{N}}. \end{aligned}$$

If $\bar{c}_{\mathcal{N}} \geq 0$ and $x_{\mathcal{N}} \geq 0$, no feasible point has value below the current basic value.

The Two Checks for an Optimal Basis

A basis is optimal when it passes both tests:

$$\boxed{B^{-1}b \geq 0} \qquad \text{and} \qquad \boxed{\bar{c}_{\mathcal{N}} \geq 0}.$$

Primal feasibility: the associated basic solution lies in the feasible set.

Optimality: no nonbasic direction has a negative objective slope.

Keep the questions separate

A nonsingular basis need not be feasible, and a feasible basis need not be optimal.

A Negative Reduced Cost Chooses an Entering Variable

Suppose a nonbasic x_j has $\bar{c}_j < 0$. Let

$$u = B^{-1}A_j, \qquad d_{\mathcal{B}} = -u, \qquad d_j = 1.$$

Along the candidate path,

$$x_{\mathcal{B}}(\theta) = x_{\mathcal{B}} - \theta u, \qquad c^\top x(\theta) = c^\top x + \theta \bar{c}_j.$$

Variable x_j **enters** the basis if the ratio test permits the move.

The Maximum Feasible Step

For each component with $u_i > 0$, nonnegativity requires

$$x_{\mathcal{B},i} - \theta u_i \geq 0 \quad \Longrightarrow \quad \theta \leq \frac{x_{\mathcal{B},i}}{u_i}.$$

Thus

$$\boxed{\theta^* = \min_{i:u_i>0} \frac{x_{\mathcal{B},i}}{u_i}}.$$

An index attaining the minimum identifies a basic variable that reaches zero and may **leave** the basis.

When No Variable Blocks the Move

If $u \leq 0$, then $d_{\mathcal{B}} = -u \geq 0$, so

$$x + \theta d \geq 0 \qquad \text{for every } \theta \geq 0.$$

Moreover,

$$Ad = 0, \qquad d \geq 0, \qquad c^\top d = \bar{c}_j < 0.$$

Unboundedness certificate

$x + \theta d$ stays feasible while its objective tends to $-\infty$. The vector d certifies that the minimization problem is unbounded.

1. Start with a feasible basis B and $x_{\mathcal{B}} = B^{-1}b$.
2. Solve $B^\top y = c_{\mathcal{B}}$ and price the nonbasic columns:

$$\bar{c}_j = c_j - y^\top A_j.$$

3. If every $\bar{c}_j \geq 0$, stop: the current basis is optimal.
4. Otherwise choose an entering j with $\bar{c}_j < 0$ and solve $Bu = A_j$.

One Simplex Iteration: State 2 of 2

1. Start with a feasible basis B and $x_{\mathcal{B}} = B^{-1}b$.
2. Price the nonbasic columns; stop if every $\bar{c}_j \geq 0$.
3. Choose an entering j with $\bar{c}_j < 0$; solve $Bu = A_j$.
4. If $u \leq 0$, return the unbounded direction d.
5. Otherwise compute $\theta^* = \min_{i : u_i > 0} x_{\mathcal{B},i}/u_i$.
6. Choose a minimizing row, exchange the entering and leaving columns, and update the basic solution.

Worked Example

$$\begin{aligned}
\min \quad & -10x_1 - 12x_2 - 12x_3 \\
\text{s.t.} \quad & x_1 + 2x_2 + 2x_3 + x_4 = 20, \\
& 2x_1 + x_2 + 2x_3 + x_5 = 20, \\
& 2x_1 + 2x_2 + x_3 + x_6 = 20, \\
& x \geq 0.
\end{aligned}$$

The slack basis $B_0 = [A_4 \ A_5 \ A_6] = I$ gives

$$x^{(0)} = (0, 0, 0, 20, 20, 20), \qquad c^\top x^{(0)} = 0.$$

Your Turn: Choose a First Pivot

At the slack basis, $c_{\mathcal{B}} = 0$, so

$$\bar{c} = (-10, -12, -12, 0, 0, 0).$$

1. Which nonbasic variables are eligible to enter?
2. Which variable would the most-negative rule choose after breaking the tie by smaller index?
3. May a different eligible variable enter under a valid simplex rule?

For this worked trace

We deliberately choose the eligible variable x_1. This is valid, but it is not the most-negative choice; that rule would select x_2.

First Pivot: Direction and Ratio Test

For entering variable x_1,

$$u = B_0^{-1} A_1 = A_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

The ratio test gives

$$\theta^* = \min \left\{ \frac{20}{1}, \frac{20}{2}, \frac{20}{2} \right\} = 10.$$

Rows x_5 and x_6 tie. Choosing the smaller variable index makes x_5 leave and x_1 enter.

After the First Pivot

The updated point is

$$x^{(1)} = (10, 0, 0, 10, 0, 0), \qquad c^\top x^{(1)} = -100.$$

The new basis is

$$B_1 = [A_4 \ A_1 \ A_6].$$

A degenerate arrival

x_6 remains basic at value zero because two ratios tied. The point is degenerate even though the step length was positive.

At B_1, both x_2 and x_3 still have negative reduced costs.

The Final Dictionary

After valid pivots, use the row order x_3, x_1, x_2. The dictionary is $x_{\mathcal{B}} = B_*^{-1}b - B_*^{-1}A_{\mathcal{N}}x_{\mathcal{N}}$, with nonbasic slacks x_4, x_5, x_6:

	value	x_4	x_5	x_6
x_3	4	$-\frac{2}{5}$	$-\frac{2}{5}$	$\frac{3}{5}$
x_1	4	$\frac{3}{5}$	$-\frac{2}{5}$	$-\frac{2}{5}$
x_2	4	$-\frac{2}{5}$	$\frac{3}{5}$	$-\frac{2}{5}$

Thus the associated basic feasible solution is

$$x^* = (4, 4, 4, 0, 0, 0).$$

Verify the Final Basis

For $B_* = [A_3 \ A_1 \ A_2]$,

$$B_*^{-1} b = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} \geq 0.$$

The nonbasic reduced costs are

$$\bar{c}_4 = \frac{18}{5}, \qquad \bar{c}_5 = \frac{8}{5}, \qquad \bar{c}_6 = \frac{8}{5}.$$

Therefore B_* is optimal and

$$c^\top x^* = -136.$$

Both the primal-feasibility and reduced-cost checks pass.

Why Simplex Terminates Without Degeneracy

Assume every basic feasible solution is nondegenerate.

- Every finite pivot has $\theta^* > 0$ and $\bar{c}_j < 0$, so the objective strictly decreases.
- A basic feasible solution cannot be visited twice.
- There are only finitely many bases.

Hence simplex terminates after finitely many pivots with either an optimal basis or an unboundedness certificate.

What the assumption buys

Strict improvement, not merely a change of basis, drives the finite argument.

Degeneracy and Cycling: A Preview

At a degenerate basic feasible solution:

- the minimum ratio can be zero;
- the basis can change while the point and objective stay fixed;
- a sequence of such pivots can revisit a basis and cycle.

Pivot-rule precision

Choosing the smallest index only for a leaving tie is not enough. Bland's anti-cycling rule selects the smallest eligible index for both entering and leaving choices.

We will study degeneracy and anti-cycling rules in the later simplex meetings.

Implementation Preview and Takeaway

Full tableau

Stores $B^{-1}b$ and $B^{-1}A$; pivots are explicit row operations.

Revised simplex

Stores a basis factorization, computes only needed solves, and exploits sparsity. It does not repeatedly form B^{-1}.

The complete logic

$$\text{feasible basis} \to \text{price} \to \text{direction}$$
$$\to \text{ratio test} \to \text{pivot or certificate}$$